# INVERTING A MULTIPLICATIVE SET OF NONZERODIVISORS DOES NOT CHANGE THE TOTAL RING OF FRACTIONS

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*Allowed standard prerequisites (editorial).*

- A fraction with nonzerodivisor numerator. Used inside the counted proof at lines 4817-4820. Shows that a nonzerodivisor of S^{-1}R written as r/f has nonzerodivisor numerator r in R.
- The total ring of fractions as a localisation at the nonzerodivisors. Definition Q(R) = (nonzerodivisors)^{-1}R, used in the statement and in the comparison with Q(S^{-1}R). Identifies the two objects that the lemma proves isomorphic and identifies the isomorphism on fractions.

**Lemma 0.1.** *Let $R$ be a ring. Let $S \subset R$ be a multiplicative subset consisting of nonzerodivisors. Then $Q(R) \cong Q(S^{-1}R)$. In particular $Q(R) \cong Q(Q(R))$.*

*Proof.* If $x \in S^{-1}R$ is a nonzerodivisor, and $x = r/f$ for some $r \in R$, $f \in S$, then $r$ is a nonzerodivisor in $R$. Whence the lemma. □

*Date*: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.